[This question paper contains 4+1 printed pages]

Roll No. | | | | | | | | | | | |

S. No. of Question Paper : 2816

Unique Paper Code : 32351403 GC-4

Name of the Paper : C10-Ring Theory and Linear Algebra-I

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any *three* parts from each question.

All questions are compulsory.

(*a*) Let R be a ring. Prove that the center of R is a subring of R. Is Z_6 a subring of Z_{12} ? 3+2=5

(*b*) Show that a finite integral domain is a field. Give an example of the same. Also, give an example of an infinite integral domain which is not a field. 3+1+1=5

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(c) Let n be an integer with decimal representation :

$$a_k\, a_{k-1} \ldots a_1\, a_0$$

Prove that n is divisible by 9 if and only if :

$$a_k + a_{k-1} + \ldots + a_1 + a_0$$

is divisible by 9. 5

(d) Show that a non-zero idempotent cannot be nilpotent. Find all the idempotent and nilpotent elements of the ring $\mathbf{Z}_4$. 2+3=5

2. (a) How many elements are in $R = \mathbf{Z}[i]/\langle 2-i \rangle$. Give reasons for your answer. 5

(b) Let R be the ring of continuous functions from **R** to **R**. Show that :

$$A = \{f \in R \mid f(0) = 0\}$$

is a maximal ideal of R. 5

(c) Define a principal ideal domain and show that **Z** is a principal ideal domain. 1+4=5

(d) Determine all ring homomorphisms from $\mathbf{Z}_{12}$ to $\mathbf{Z}_{30}$. 5

(a) Prove that union of two subspaces of a vector space is a subspace if and only if one of the subspace is contained in the other. 5

(b) Show that for a subset W of a vector space V, W = span (W) if and only if W is a subspace of V. 5

(c) Check whether the subset :

S = {1, 1, 2, 4), (2, −1, −5, 2), (1, −1, −4, 0), (2, 1, 1, 6) of $\mathbf{R}^4$ is linearly dependent or linearly independent. 5

(d) Suppose that V be a vector space over a field F and let S = $\{x_1, x_2, ..., x_n\}$ be a linearly independent subset of V. Further, let $x \in$ V be such that $x \notin$ S then :

$$S_1 = \{x_1, x_1, x_2, ..., x_n\}$$

is linearly dependent if and only if $x \in$ span(S). 5

(a) Let S be a finite subset of vector space V such that V = span(S). Then, prove that there exists a subset T of S such that T is a basis of V. 5

(*b*) Let $\beta = \{v_1, v_2, ..., v_n\}$ be a subset of vector space V over F. Then , β forms a basis for V if and only if every element of V can be expressed uniquely as a linear combination of elements of β. 5

(*c*) Find bases for the following subspace of $\mathbf{R}^5$:

$$W_1 = \{(a_1, a_2, a_3, a_4, a_5\} \in \mathbf{R}^5 : a_1 - a_3 - a_4 = 0\} \text{ and}$$

$$W_2 = \{(a_1, a_2, a_3, a_4, a_5) \in \mathbf{R}^5 : a_2 = a_3 = a_4 \text{ and } a_1 + a_5 = 0\}.$$

What are the dimensions of W_1 and W_2 ? 2,2,1

(*d*) Extend the set $S = \{1, 1, 0)\}$ to form two different bases of $\mathbf{R}^3$. 2.5,2.5

5. (*a*) Let V and W be vector spaces and $T : V \to W$ be a linear transformation.

If $\beta = \{v_1, v_2 ..., v_n\}$ is a basis for V, then prove that : 5

$$R(T) = \text{span}\,(T(\beta)) = \text{span}\,(\{T(v_1), T(v_2), .., T(v_n)\})$$

(*b*) Let $T : \mathbf{R}^2 \to \mathbf{R}^3$ be defined by :

$$T(x_1, x_2) = (x_1 - x_2, x_1, 2x_1 + x_2) .$$

Let β be the standard ordered basis for $\mathbf{R}^2$ and :

$$\gamma = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$$

Compute $[T]_\beta^\gamma$. 5

(c) Let V, W and Z be vector spaces and let $T : V \to W$ and $U : W \to Z$ be linear transformations :

(i) Prove that if UT is one-to-one, then T is one-to-one.

(ii) Prove that if UT is onto, then U is onto. 2.5.,2.5.

(d) Let :

$$A = \begin{pmatrix} 13 & 1 & 4 \\ 1 & 13 & 4 \\ 4 & 4 & 10 \end{pmatrix} \text{ and } \beta = \left\{ \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

be an ordered basis of $\mathbf{R}^3$. Then find $[L_A]_\beta$ and also find an invertible matrix Q such that $[L_A]_\beta = Q^{-1} AQ$, where L_A is a left-multiplication transformation. 5